

Clinical Immunogenicity Evaluation of Eptinezumab, a Therapeutic Humanized Monoclonal Antibody Targeting Calcitonin Gene-Related Peptide (CGRP) for the Preventive Treatment of Migraine.

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Eptinezumab is a humanized monoclonal antibody that selectively binds calcitonin gene-related peptide and is indicated for the preventive treatment of migraine in adults. This analysis characterizes the immunogenic profile of eptinezumab using data from clinical trials of eptinezumab for migraine prevention. Immunogenicity data were collected from five studies that included 2076 patients with episodic or chronic migraine treated with eptinezumab at dose levels ranging from 10 to 1000 mg, administered intravenously for up to 4 doses at 12-week intervals. Anti-drug antibody (ADA) results were available from 2074 of these patients. Four studies were randomized, double-blind, placebo-controlled trials with ADA monitoring for up to 56 weeks; one was a 2-year, open-label, phase 3 safety study with ADA monitoring for 104 weeks. Patients who had a confirmed ADA-positive result at the end-of-study visit were monitored for up to 6 additional months. Development of ADA and neutralizing antibodies (NABs) were evaluated to explore three key areas of potential impact: pharmacokinetic exposure profile (eptinezumab trough plasma concentrations), efficacy (change in monthly migraine days), and safety (rates of treatment-emergent adverse events). These studies included methods designed to capture the dynamics of a potential humoral immune response to eptinezumab treatment, and descriptive analyses were applied to interpret the relationship of ADA signals to drug exposure, efficacy, and safety. Pooled across the five clinical trials, treatment-emergent ADAs and NABs occurred in 15.8 and 6.2% of eptinezumab-treated patients, respectively. Highly consistent profiles were observed across all studies, with initial onset of detectable ADA observed at the week 8 measurement and maximal ADA frequency and titer observed at week 24, regardless of eptinezumab dose level or number of doses. After 24 weeks, the ADA and NAb titers steadily declined despite additional doses of eptinezumab. Collectively, these integrated analyses did not demonstrate any clinically meaningful impact from ADA occurring after treatment with eptinezumab. The ADA profiles were low titer and transient, with the incidence and magnitude of ADA or NAb responses declining after week 24. Development of ADAs and NABs did not impact the efficacy and safety profiles of eptinezumab.

Eptinezumab for the preventive treatment of episodic and chronic migraine: a narrative review.

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Eptinezumab, a monoclonal antibody that targets calcitonin gene-related peptide (CGRP), was recently approved in Europe for the prophylactic treatment of migraine in adults who have at least four migraine days a month. Eptinezumab is administered by intravenous infusion every 12 weeks. During recent months, a considerable amount of evidence from eptinezumab trials has been published. The aim of this review is to describe the existing evidence on the tolerability, safety and efficacy of eptinezumab in patients with migraine. Data from randomized (PROMISE-1, PROMISE-2, RELIEF and DELIVER) and open-label (PREVAIL) phase 3 clinical trials have demonstrated the favorable effect of eptinezumab in migraine symptoms from first day of treatment. These studies showed that eptinezumab results in an overall reduction in mean monthly migraine days (MMDs), increases in the $\geq 50\%$ and $\geq 75\%$ migraine responder rates (MRRs) and improvements in patient-reported outcome measures in both patients with episodic migraine (EM) and with chronic migraine (CM), including patients who failed previous preventive

treatments. The RELIEF trial also showed that eptinezumab, within 2 h of administration, reduced headache pain, migraine-associated symptoms and acute medication use when administered during a migraine attack. Eptinezumab benefits manifested as early as day 1 after dosing and with the subsequent doses lasted up to at least 2 years. Treatment-emergent adverse events reported by $\geq 2\%$ of patients included upper respiratory tract infection and fatigue. Current evidence demonstrates that eptinezumab has a potent, fast-acting, sustained migraine preventive effect in patients with EM and CM. Eptinezumab has also shown to be well tolerated, supporting its use in the treatment of patients with migraine and inclusion in the current migraine therapeutic options.

Eptinezumab for the Prevention of Migraine: Clinical Utility, Patient Preferences and Selection - A Narrative Review.

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The new Calcitonin Gene-Related Peptide (CGRP)-targeted therapies have proven high efficacy and tolerability in episodic and chronic migraine. Eptinezumab is a humanized monoclonal antibody that selectively binds CGRP with high affinity. Eptinezumab was approved by the Food and Drug Administration on February 21st, 2020, for the preventive treatment of migraine in adults. It is administered intravenously over 30 minutes with a standard dose of 100 mg and has a T-max of 30 minutes-1 hour and a half-life of 27 days. These pharmacological properties allow for a very rapid onset of effect and a quarterly administration. It is the first time that a preventive treatment for migraine can be offered as an intravenous administration. As the range of therapeutic possibilities in migraine is expanding, the treatment process must include common decision-making, where physicians should explain in detail to patients the different characteristics of treatment options beyond efficacy and side effects. Patients can now express a preference on a range of opportunities: pharmacological versus non-pharmacological approaches, route of administration, frequency of administration, efficacy, rapidity, side effects, costs, the possibility of titration or dosing, and durability of effectiveness at suspension. Also, patient preferences can be influenced by age, country, migraine severity, and earlier experience with CGRP-targeted therapies. Besides, adherence may be influenced by several factors, including route and the schedule of administration. This narrative review describes a new perspective from the patient's point of view. Clinicians should ally with patients to select treatments that meet each patient's needs and thus apply a tailored approach, addressing not only headaches. In this way, physicians would care for the patients globally and stand out their preferences on different aspects of treatment. Besides, healthcare professionals shall be aware that patients' beliefs about therapies are subject to change with increasing experience with new therapeutic approaches.

Eptinezumab: A calcitonin gene-related peptide monoclonal antibody infusion for migraine prevention.

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This article seeks to analyze the clinical trials concerning the newly approved eptinezumab to assess its efficacy, safety, and application to current clinical practice. The Institute of Health US National Library of Medicine Clinical Trials, PubMed, and Cochrane Library databases were searched for relevant abstracts, journal articles, and other published sources. Search terms included eptinezumab, Vyepti®, and ALD403.

Relevant English-language articles were evaluated and included in the narrative. Two randomized controlled trials compared quarterly infusions of eptinezumab 100 mg, eptinezumab 300 mg, and placebo in chronic and episodic migraine sufferers. In episodic migraine, eptinezumab resulted in a reduction of approximately 4 monthly migraine days, which was significant compared to placebo. In chronic migraine, eptinezumab reduced monthly migraine days by approximately 8 days, also significant compared to placebo. More patients who received eptinezumab experienced at least 75% reduction in monthly migraine days compared to placebo, resulting in a number needed to treat as low as 6, depending on the study population and the dose. The preventive impact was noticed day one post-infusion. The most common treatment-emergent adverse events were nausea and fatigue, and there was a low incidence of hypersensitivity or study withdrawal. Eptinezumab is the fourth Calcitonin Gene-related Peptide monoclonal antibody to receive Federal Drug Administration approval. Its delivery as a quarterly infusion sets it apart from the other agents in this class. As an infusion, eptinezumab has a quick onset of action that may prove especially beneficial to those with severe or refractory episodic or chronic migraines, despite the perceived increased direct and indirect cost of an infusion.

Rapid resolution of migraine symptoms after initiating the preventive treatment eptinezumab during a migraine attack: results from the randomized RELIEF trial.

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Eptinezumab is an anti-calcitonin gene-related peptide (CGRP) monoclonal antibody approved for the preventive treatment of migraine. In the phase 3 RELIEF study, eptinezumab resulted in shorter time to headache pain freedom and time to absence of most bothersome symptom (MBS; including nausea, photophobia, or phonophobia) compared with placebo when administered during a migraine attack. The objective of this exploratory analysis was to examine the earliest time points that eptinezumab separated from placebo ($P < .05$) on headache- and migraine-associated symptoms when administered during a migraine attack. RELIEF, a multicenter, parallel-group, double-blind trial, occurred from November 7, 2019, through July 8, 2020. Adults considered candidates for preventive treatment were randomized to eptinezumab 100 mg ($N = 238$) or placebo ($N = 242$) administered intravenously over 30 min within 1-6 h of migraine onset. Outcome measures included headache pain freedom/relief and absence of MBS, patient's choice of photophobia, phonophobia, or nausea, at regular intervals from 0.5 to 48 h after infusion start. Censoring was applied at time of acute rescue medication use. At hour 1, more eptinezumab-treated patients achieved headache pain freedom (9.7%), headache pain relief (38.7%), and absence of MBS (33.2%) versus placebo (4.1%, 26.9%, and 22.1%, respectively; $P < .05$ all), with separation from placebo ($P < .05$) through hour 48. Eptinezumab separated from placebo ($P < .05$) at hour 1 in absence-of-photophobia (29.4% vs 17.0%) and absence-of-phonophobia (41.2% vs 27.2%) and through hour 48. Initial separation from placebo ($P < .05$) in absence-of-nausea occurred at end-of-infusion (0.5 h; 36.7% vs 25.4%, respectively). Preventive treatment with eptinezumab initiated during a migraine attack resulted in more patients achieving headache pain freedom/relief and absence of MBS, with separation from placebo ($P < .05$) as early as 0.5-1 h following the start of infusion. Rapid resolution of headache- and migraine-associated symptoms by a peripherally acting, intravenously administered antibody suggest a peripheral site of pharmacological action for CGRP blockade. ClinicalTrials.gov Identifier: NCT04152083 .

Eptinezumab-jjmr, a humanized monoclonal specific to Calcitonin Gene Related Peptide,

for the preventive treatment of migraine in adults.

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Migraines are prevalent and cause significant morbidity, decline in quality of life and healthcare costs universally. Treatment options are varied, but efficacy is limited. This review centers on Eptinezumab-jjmr, a humanized monoclonal specific to CGRP for the prevention of migraines in adults. Herein presented are the science and mechanism of action, indication and clinical evidence for use. Migraines are severe, recurrent headaches, which are either episodic or chronic in nature. The pain is severe, often accompanied by co-morbid symptoms, such as photophobia, phonophobia, nausea and emesis, and is limiting in nature. It is a prevalent disorder that causes significant, worldwide disability, morbidity, suffering, and costs. The pathophysiology of migraines is actively studied, though recent research points to an initiating event causing migraine generation, that is then propagated by other brain regions, a significant one being the trigeminocervical complex. This is driven by biochemical transmitters, chiefly CGRP. This discovery led to the development of CGRP-targeting drugs, including gepants (small molecular antagonists) and anti-CGRP antibodies, such as Eptinezumab-jjmr. Traditional therapy includes preventative and abortive treatment; however, adherence with preventative treatment has been historically poor, and certain types of abortive therapy carry risks and side effects that preclude them from a large patient population. Moreover, traditional therapy often falls short in migraine therapy. CGRP antagonist, including Eptinezumab, aims to cover the gaps in migraine therapy. We present here evidence to support the safe and effective use of Eptinezumab for the prevention of migraines. Migraines are a prevalent primary headache disorder causing significant morbidity worldwide. Traditional abortive and preventative treatments fall short for many patients. Eptinezumab is part of new generation of CGRP-targeting medications and has shown significant evidence to support its use for the prevention of migraines. Further research is required to properly compare eptinezumab with existing pharmacotherapy and update guidelines on the appropriate combinations of therapies that are not available and the correct patient selection for each.

Persistence to anti-CGRP monoclonal antibodies and onabotulinumtoxinA among patients with migraine: a retrospective cohort study.

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To date, real-world evidence on persistence to anti-calcitonin gene-related peptide (anti-CGRP) monoclonal antibodies (mAbs) or onabotulinumtoxinA have excluded eptinezumab. This retrospective cohort study was performed to compare treatment persistency among patients with migraine on anti-CGRP mAbs (erenumab, fremanezumab, galcanezumab, or eptinezumab) or onabotulinumtoxinA. This retrospective study used IQVIA PharmMetrics data. Adult patients with migraine treated with an anti-CGRP mAb or onabotulinumtoxinA who had 12 months of continuous insurance enrollment before starting treatment were included. A "most recent treatment episode" analysis was used in which the most recent episode was defined as the latest treatment period with the same drug (anti-CGRP mAb or onabotulinumtoxinA) without a ≥ 15 -day gap in medication supply on/after June 25, 2020, to December 31, 2021. Patients were indexed at the start of their most recent episode. Patients were considered non-persistent and discontinued the therapy associated with their most recent episode if there was ≥ 15 -day gap in medication supply. A Cox proportional-hazards model estimated the discontinuation hazard between treatments. The gap periods and cohort definition were varied in sensitivity analyses. The study included 66,576 patients (median age 46 years, 88.6% female). More eptinezumab-treated patients had chronic migraine (727/1074), ≥ 3 previous acute (323/1074) or preventive (333/1074)

therapies, and more prior treatment episodes (3) than other treatment groups. Based on a 15-day treatment gap, patients on subcutaneous anti-CGRP mAbs had a 32% (95% CI: 1.19, 1.49; erenumab), 42% (95% CI: 1.27, 1.61; galcanezumab), and 58% (95% CI: 1.42, 1.80; fremanezumab) higher discontinuation hazard than those receiving eptinezumab, with this relationship attenuated, but still statistically significant based on 30-day and 60-day treatment gaps. There was no significant difference in the discontinuation hazard between eptinezumab and onabotulinumtoxinA. Based on a 15-day treatment gap among patients who newly initiated therapy, the discontinuation hazard of subcutaneous anti-CGRP mAbs remained significantly higher compared to eptinezumab and onabotulinumtoxinA. Patients treated with eptinezumab demonstrated persistency that was higher than subcutaneous anti-CGRP mAbs and similar to onabotulinumtoxinA.

Commentary: Constipation caused by anti-calcitonin gene-related peptide migraine therapeutics explained by antagonism of calcitonin gene-related peptide's motor-stimulating and prosecretory function in the intestine.

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Calcitonin gene-related peptide (CGRP) pathway-targeted treatments have been shown to be efficacious in the prevention of episodic and chronic migraine. Currently approved therapies include monoclonal antibodies (mAbs) that target CGRP (eptinezumab, fremanezumab, and galcanezumab) and the CGRP receptor (erenumab), and small molecule CGRP receptor antagonists (atogepant and rimegepant). While CGRP pathway-targeted treatments are generally well-tolerated, in a review article by Holzer and Holzer-Petsche published in the January 2022 issue of *Frontiers in Physiology* the authors discussed the role of the CGRP pathway in gastrointestinal physiology, with a specific focus on constipation associated with the use of CGRP pathway-targeted treatments. The authors state that real-world surveys have shown constipation to be a "major adverse event" reported in "more than 50% of patients treated with erenumab, fremanezumab or galcanezumab." As described in the current commentary, the limited data from the cited references in the review article by Holzer and Holzer-Petsche do not support that statement.

An observational study on monoclonal antibodies against calcitonin-gene-related peptide and its receptor.

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Based on their pharmacological target, two classes of calcitonin-gene-related peptide (CGRP) monoclonal antibodies (mAbs) have been identified: antibodies against the CGRP ligand-galcanezumab, fremanezumab, eptinezumab-and antibodies against the CGRP receptor (CGRP-R), erenumab. The aim of the present study was to compare anti-CGRP versus anti-CGRP-R mAbs in patients with high frequency episodic and chronic migraine. All patients on monthly treatment with anti-CGRP mAbs with an available 6 months' follow-up at January 2022 were included. Data on efficacy outcome were collected following one (T1), three (T3) and six (T6) months of treatment, and included monthly headache/migraine days, the Migraine Disability Assessment Scale (MIDAS) and Headache Impact Test 6 (HIT-6) scores, pain intensity, analgesics consumption and response rates (>50% headache days reduction compared to baseline). In all, 152 patients were enrolled, of whom 68 were in treatment with anti-CGRP mAbs (49

galcanezumab, 19 fremanezumab) and 84 with the anti-CGRP-R (erenumab). MIDAS scores were significantly lower in the anti-CGRP group at T1 and T3 (respectively $p < 0.02$ and $p < 0.03$) as well as the number of mean migraine days at T3 ($p < 0.01$). At T3 and T6 outcome measures were comparable, although a significantly higher percentage of super-responders was found in the anti-CGRP group (respectively $p < 0.04$ and $p < 0.05$), with a similar overall percentage of responders. The present study on a real-world sample confirms the beneficial effect of both anti-CGRP and anti-CGRP-R mAbs, with a more favorable outcome for anti-CGRP antibodies.

Eptinezumab for prevention of chronic migraine: A randomized phase 2b clinical trial.

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Calcitonin gene-related peptide plays an important role in migraine pathophysiology. We evaluated eptinezumab, an intravenous (IV) anti-calcitonin gene-related peptide monoclonal antibody, for the prevention of chronic migraine. To determine the safety, tolerability, and effectiveness of four dose levels of eptinezumab and to inform the phase 3 development program. This was a phase 2b, parallel-group, double-blind, randomized, placebo-controlled, dose-ranging clinical trial. Men and women ($N = 616$) aged 18-55 years were included if they had a diagnosis of chronic migraine, with onset at age ≤ 35 years and history of chronic migraine ≥ 1 year. During the 28-day screening period, patients must have had ≥ 15 headache days, including ≥ 8 migraine days, with ≥ 5 migraine attacks as recorded in the electronic diary. Patients were assigned in a 1:1:1:1 ratio to eptinezumab 300, 100, 30, 10 mg or placebo, administered as a single IV infusion. The primary endpoint was the percentage of patients with a $\geq 75\%$ decrease in monthly migraine days over weeks 1-12 compared with the 28-day screening period. The $\geq 75\%$ migraine responder rates over weeks 1-12 for eptinezumab 300, 100, 30, and 10 mg were 33.3%, 31.4%, 28.2%, and 26.8%, respectively, versus 20.7% for placebo ($p = 0.033, 0.072, 0.201, 0.294$ vs. placebo). Secondary efficacy endpoints (e.g. $\geq 50\%$ responder rate, change from baseline in frequency of migraine/headache days, and percentage of severe migraines) had results favoring the three higher eptinezumab doses versus placebo. Eptinezumab was well tolerated and adverse event rates were similar to placebo. The results of this trial demonstrate that eptinezumab appears effective and well-tolerated for the preventive treatment of chronic migraine and justifies the conduct of pivotal phase 3 trials for migraine prevention. ClinicalTrials.gov identifier: NCT02275117.